

Mechanics, Hydrodynamics & Angular Motion Master Paper

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Subject: Physics (Unit 02: Mechanics & Fluid Dynamics)

Target Year: 2026 A/L Exam

Format: Structured / Essay Level

Question 01

Rotational Dynamics & Energy

A uniform solid sphere of mass M and radius R is placed on an inclined plane of inclination angle θ . The coefficient of static friction between the sphere and the inclined plane is μ_s , and the coefficient of kinetic friction is μ_k .

1. Derive the minimum value of μ_s required for the sphere to roll purely without slipping down the incline.
2. If μ_s is reduced to half of this minimum value, the sphere rolls with slipping.
 - Calculate the linear acceleration a and angular acceleration α of the sphere down the incline.
 - Calculate the velocity v of the center of mass when the sphere has traveled a distance S down the incline from rest.
 - Show that the work done against friction over this distance S is equal to the mechanical energy lost by the system.

(Moment of inertia of a uniform solid sphere about its center of mass: $I = \frac{2}{5}MR^2$)

Question 02

Angular Momentum & Conservation

A uniform thin horizontal rod of mass M and length L is free to rotate in a horizontal plane about a frictionless vertical axis passing through its center O . Initially, two small sleeves, each of mass m , are attached to the rod at a distance $r_0 = \frac{L}{4}$ on opposite sides of O . The assembly rotates with an initial angular velocity ω_0 .

1. Write down the total initial angular momentum L_i and initial rotational kinetic energy K_i of the system in terms of M , m , L , and ω_0 .
2. An internal mechanism releases the sleeves, allowing them to slide freely along the frictionless rod towards its outer ends. Find the final angular speed ω_f of the system just as the sleeves reach the ends of the rod.
3. Calculate the change in kinetic energy ΔK of the system. Explain physically where this energy change originates.
4. If the sleeves fly off the ends of the rod simultaneously, determine the velocity vector of each sleeve immediately after detachment relative to a ground observer.

(Moment of inertia of a uniform thin rod about its center of mass: $I_{rod} = \frac{1}{12}ML^2$)

Question 03

Fluid Dynamics & Bernoulli's Equation

A large enclosed cylindrical water tank filled with water of density ρ has a cross-sectional area A . The air pressure in the sealed space above the water surface is maintained at a constant gauge pressure P_0 . Near the base of the tank, a horizontal pipe of varying cross-section is connected. The pipe tapers from an area A_1 to a narrow throat of area A_2 ($A_2 < A_1 \ll A$) and opens out to the atmosphere at atmospheric pressure P_{atm} .

1. Assuming streamline flow of an incompressible, non-viscous fluid, derive an expression for the efflux speed v_2 of water exiting the throat into the atmosphere when the water height in the tank is h above the pipe level.
2. Find the gauge pressure P_1 inside the wider section of the pipe (area A_1) as a function of v_2 , A_1 , and A_2 .
3. A vertical U-tube manometer containing mercury (density ρ_m) is connected between section 1 (A_1) and section 2 (A_2). Show that the difference in mercury level Δh in the manometer is given by:

$$\Delta h = \left(\rho \cdot v_2^2 / 2g (\rho_m - \rho) \right) \left[1 - (A_2/A_1)^2 \right]$$

Question 04

Hydrostatics & Fluid Pressure

A vertical cylindrical vessel of cross-sectional area A contains two immiscible liquids: an upper layer of liquid 1 of density ρ_1 and depth h_1 , and a lower layer of liquid 2 of density ρ_2 ($\rho_2 > \rho_1$) and depth h_2 . A solid uniform cylinder of mass m , length L , and cross-sectional area a ($a < A$) floats vertically in the vessel such that it penetrates into both liquid layers.

1. Derive an expression for the total gauge pressure at the bottom surface of the floating solid cylinder.
2. Find the length x of the cylinder submerged in liquid 2 in terms of m , a , L , ρ_1 , and ρ_2 .
3. Determine the maximum density ρ_s of the cylinder material such that it remains floating with its upper end entirely inside liquid 1 (not emerging into the air above).
4. If the floating cylinder is pushed down slightly by a distance y and released, prove that it executes Simple Harmonic Motion (SHM) and find its angular frequency ω of oscillation.

Question 05

Projectile Motion & Vector Bounds

A particle is projected from a horizontal surface with an initial velocity u at an angle θ to the horizontal. A vertical boundary wall of height H is situated at a horizontal distance D from the point of projection.

1. Show that the required angle of projection θ for the particle to clear the wall must satisfy:

$$\tan^2 \theta - \left(\frac{2u^2}{gD} \right) \tan \theta + \left(1 + \frac{2gH}{u^2} \right) \leq 0$$

2. Find the minimum initial speed u_{min} required to clear the wall of height H at distance D .
3. If $u = \sqrt{2g(H + D)}$, determine the range of values for θ such that the particle clears the wall.

Question 06

Impulse & 2D Oblique Collisions

Two uniform smooth spheres **A** and **B**, of equal radii but masses **m** and **$2m$** respectively, lie on a smooth horizontal surface. Sphere **A** moves with speed **u** and impacts sphere **B**, which is initially at rest. At the instant of impact, the line joining their centers makes an angle of **30°** with the original direction of motion of sphere **A**.

The coefficient of restitution between the spheres is **$e = \frac{1}{2}$** .

1. Find the components of the velocity of sphere **A** parallel and perpendicular to the line of impact immediately after the collision.
2. Determine the angle through which the path of sphere **A** is deflected after the impact.
3. Calculate the fractional loss of kinetic energy during the collision.